

## Tabla original (F1, folds=3, Tabla 6 de la tesis)

Cuadro 1: AUC-ROC y precisión en el decil de mayor riesgo, con y sin DMSP-OLS, holdout temporal (train 2010→2013, test 2013→2016).

| Algoritmo              | Especificación | AUC-ROC | IC95 %         | Precision top-10 % | IC95 %         | Recall | Precision | F1    |
|------------------------|----------------|---------|----------------|--------------------|----------------|--------|-----------|-------|
| XGBoost                | A              | 0.763   | [0.761, 0.764] | 0.485              | [0.471, 0.499] | 0.710  | 0.349     | 0.467 |
| XGBoost                | A + DMSP-OLS   | 0.762   | [0.761, 0.762] | 0.484              | [0.479, 0.489] | 0.718  | 0.344     | 0.465 |
| HistGradientBoosting   | A              | 0.754   | [0.754, 0.754] | 0.491              | [0.491, 0.491] | 0.736  | 0.332     | 0.457 |
| HistGradientBoosting   | A + DMSP-OLS   | 0.755   | [0.755, 0.755] | 0.491              | [0.491, 0.491] | 0.692  | 0.345     | 0.458 |
| Logística regularizada | A              | 0.747   | [0.747, 0.747] | 0.506              | [0.506, 0.506] | 0.752  | 0.310     | 0.438 |
| Logística regularizada | A + DMSP-OLS   | 0.746   | [0.746, 0.746] | 0.500              | [0.500, 0.500] | 0.753  | 0.311     | 0.439 |
| XGBoost                | B              | 0.734   | [0.732, 0.735] | 0.488              | [0.482, 0.494] | 0.665  | 0.332     | 0.442 |
| XGBoost                | B + DMSP-OLS   | 0.732   | [0.729, 0.734] | 0.475              | [0.466, 0.484] | 0.781  | 0.302     | 0.435 |
| HistGradientBoosting   | B              | 0.734   | [0.734, 0.734] | 0.484              | [0.484, 0.484] | 0.662  | 0.326     | 0.437 |
| HistGradientBoosting   | B + DMSP-OLS   | 0.735   | [0.735, 0.735] | 0.475              | [0.475, 0.475] | 0.663  | 0.340     | 0.449 |
| Logística regularizada | B              | 0.739   | [0.739, 0.739] | 0.500              | [0.500, 0.500] | 0.697  | 0.317     | 0.435 |
| Logística regularizada | B + DMSP-OLS   | 0.739   | [0.739, 0.739] | 0.494              | [0.494, 0.494] | 0.698  | 0.315     | 0.434 |

## F-beta ( $\beta = 2$ ), folds=3

Cuadro 2: AUC-ROC y precisión en el decil de mayor riesgo, con y sin DMSP-OLS, holdout temporal (train 2010→2013, test 2013→2016). Umbral elegido por CV maximizando F-beta ( $\beta = 2$ , pesa recall el doble que precision) en vez de F1 – comparar con Tabla ??.

| Algoritmo<br>F1                 | Especificación | AUC-ROC | IC95 %         | Precision top-10 % | IC95 %         | Umbral | Recall | Precision |
|---------------------------------|----------------|---------|----------------|--------------------|----------------|--------|--------|-----------|
| XGBoost<br>0.424                | A              | 0.763   | [0.761, 0.764] | 0.485              | [0.471, 0.499] | 0.160  | 0.931  | 0.274     |
| XGBoost<br>0.431                | A + DMSP-OLS   | 0.762   | [0.761, 0.762] | 0.484              | [0.479, 0.489] | 0.164  | 0.922  | 0.281     |
| HistGradientBoosting<br>0.427   | A              | 0.754   | [0.754, 0.754] | 0.491              | [0.491, 0.491] | 0.138  | 0.907  | 0.279     |
| HistGradientBoosting<br>0.426   | A + DMSP-OLS   | 0.755   | [0.755, 0.755] | 0.491              | [0.491, 0.491] | 0.130  | 0.910  | 0.278     |
| Logística regularizada<br>0.424 | A              | 0.747   | [0.747, 0.747] | 0.506              | [0.506, 0.506] | 0.370  | 0.884  | 0.279     |
| Logística regularizada<br>0.423 | A + DMSP-OLS   | 0.746   | [0.746, 0.746] | 0.500              | [0.500, 0.500] | 0.362  | 0.890  | 0.278     |
| XGBoost<br>0.385                | B              | 0.734   | [0.732, 0.735] | 0.488              | [0.482, 0.494] | 0.156  | 0.940  | 0.242     |
| XGBoost<br>0.376                | B + DMSP-OLS   | 0.732   | [0.729, 0.734] | 0.475              | [0.466, 0.484] | 0.114  | 0.966  | 0.233     |
| HistGradientBoosting<br>0.414   | B              | 0.734   | [0.734, 0.734] | 0.484              | [0.484, 0.484] | 0.136  | 0.885  | 0.270     |
| HistGradientBoosting<br>0.417   | B + DMSP-OLS   | 0.735   | [0.735, 0.735] | 0.475              | [0.475, 0.475] | 0.122  | 0.877  | 0.274     |
| Logística regularizada<br>0.414 | B              | 0.739   | [0.739, 0.739] | 0.500              | [0.500, 0.500] | 0.348  | 0.893  | 0.270     |
| Logística regularizada<br>0.417 | B + DMSP-OLS   | 0.739   | [0.739, 0.739] | 0.494              | [0.494, 0.494] | 0.350  | 0.886  | 0.273     |

## F-beta ( $\beta = 2$ ), folds=10, iter=30

Cuadro 3: AUC-ROC y precisión en el decil de mayor riesgo, con y sin DMSP-OLS, holdout temporal (train 2010→2013, test 2013→2016). Umbral elegido por CV maximizando F-beta ( $\beta = 2$ ), con CV\_FOLDS=10 y N\_ITER\_BUSQUEDA=30 – comparar con Tabla ?? (F1, folds=3) y Tabla ?? (F-beta=2, folds=3).

| Algoritmo<br>F1                 | Especificación | AUC-ROC | IC95 %         | Precision top-10 % | IC95 %         | Umbral | Recall | Precision |
|---------------------------------|----------------|---------|----------------|--------------------|----------------|--------|--------|-----------|
| XGBoost<br>0.413                | A              | 0.767   | [0.766, 0.769] | 0.505              | [0.496, 0.514] | 0.112  | 0.949  | 0.264     |
| XGBoost<br>0.421                | A + DMSP-OLS   | 0.765   | [0.764, 0.766] | 0.501              | [0.488, 0.515] | 0.118  | 0.937  | 0.272     |
| HistGradientBoosting<br>0.432   | A              | 0.763   | [0.763, 0.763] | 0.512              | [0.512, 0.512] | 0.324  | 0.909  | 0.283     |
| HistGradientBoosting<br>0.429   | A + DMSP-OLS   | 0.763   | [0.763, 0.763] | 0.500              | [0.500, 0.500] | 0.312  | 0.912  | 0.281     |
| Logística regularizada<br>0.426 | A              | 0.747   | [0.747, 0.747] | 0.506              | [0.506, 0.506] | 0.372  | 0.883  | 0.280     |
| Logística regularizada<br>0.426 | A + DMSP-OLS   | 0.746   | [0.746, 0.746] | 0.500              | [0.500, 0.500] | 0.370  | 0.883  | 0.281     |
| XGBoost<br>0.395                | B              | 0.742   | [0.742, 0.743] | 0.495              | [0.487, 0.503] | 0.126  | 0.940  | 0.250     |
| XGBoost<br>0.399                | B + DMSP-OLS   | 0.740   | [0.740, 0.741] | 0.495              | [0.491, 0.499] | 0.128  | 0.929  | 0.254     |
| HistGradientBoosting<br>0.418   | B              | 0.735   | [0.735, 0.735] | 0.494              | [0.494, 0.494] | 0.142  | 0.879  | 0.275     |
| HistGradientBoosting<br>0.419   | B + DMSP-OLS   | 0.736   | [0.736, 0.736] | 0.487              | [0.487, 0.487] | 0.142  | 0.880  | 0.275     |
| Logística regularizada<br>0.414 | B              | 0.739   | [0.739, 0.739] | 0.500              | [0.500, 0.500] | 0.350  | 0.890  | 0.270     |
| Logística regularizada<br>0.417 | B + DMSP-OLS   | 0.739   | [0.739, 0.739] | 0.494              | [0.494, 0.494] | 0.350  | 0.888  | 0.273     |